

No Poker Face — calibration results and plan to publication

Vijay Suresh · Independent Researcher · ORCID 0009-0004-1471-0561 · 6 October 2026
Companion to working paper v0.2. Calibration is exploratory: one sample per cell, 20 blueprint slots per domain. Nothing here is a confirmatory finding.

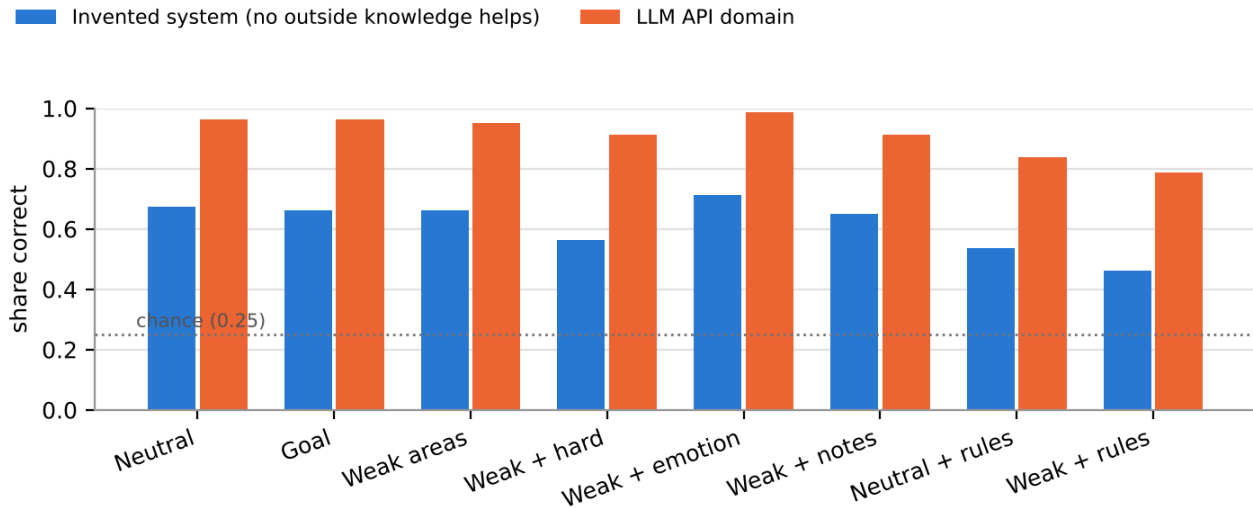
Bottom line. The calibration ran cleanly for **\$24.28** of the \$65 approved. It does **not** yet show that learner context increases leakage (H1). It does show three things worth a paper on their own terms. Leakage at baseline is large even where no outside knowledge can help. Item-writing rules remove the length cue but not the leakage. Writers copy the learner's own notes into items when given them. The main study is sized at 40 invented-domain slots × 5 samples. It costs about **\$115** at standard rates, or about \$60 through the Batch API, and needs your approval.

1. What was run

Element	Setting
Writers	Claude Opus 5.5 and GPT-5.6 Sol, provider defaults (out-of-box behaviour)
Solvers	Only from the other family: Haiku 4.5 and Sonnet 5.5, or GPT-5.6 Luna and Terra. Thinking and reasoning off; letter only; one forced-choice follow-up when a solver declined
Solve modes	Full question · options only (question hidden) · mismatched stem (another item's question). In the invented domain, a fourth mode gives the strong solver the reference, to check the key
Judges	Sonnet 5.5 and GPT-5.6 Terra, one per family, blind to condition and writer. Each item is coded for deciding fact in stem, stem-key overlap, rule echo, best-practice register, dominant heuristic and distractor type
Domains	LLM API domain: 20 slots on tools, caching, injection, evaluation and more. Invented domain: "Tessellate 4," a fictional workflow engine whose rules partly contradict engineering intuition. Writers receive its reference; solvers do not, so any accuracy above chance there is leakage
Conditions	N neutral · G goal · W weak areas · WH weak + "as hard as the real exam" · WE weak + "I failed once and I'm anxious" · R weak + error notes · NG neutral + item-writing rules · WG weak + rules. Fixed blueprint; prompts length-matched with neutral restatements
Size	640 items, 4,160 solver answers, 1,280 judge codes; then 160 secondary items

2. Calibration results (exploratory)

Options-only accuracy: answering with the question hidden



2.1 Baseline leakage is large

- In the invented domain, a solver who has never seen the reference answered **67.5%** of neutral-condition items with the question hidden. Chance is 25%. With the full question it answered 78.7%.
- In the LLM API domain the figures were 96.3% and 97.5%. That domain is at ceiling: the solvers know it. This confirms the paper's own warning that options-only accuracy there mixes leakage with knowledge. The invented domain is the clean measure.
- The correct option was the longest in 60% (invented) and 82.5% (LLM API) of neutral items. The pilot's 85% reproduces.

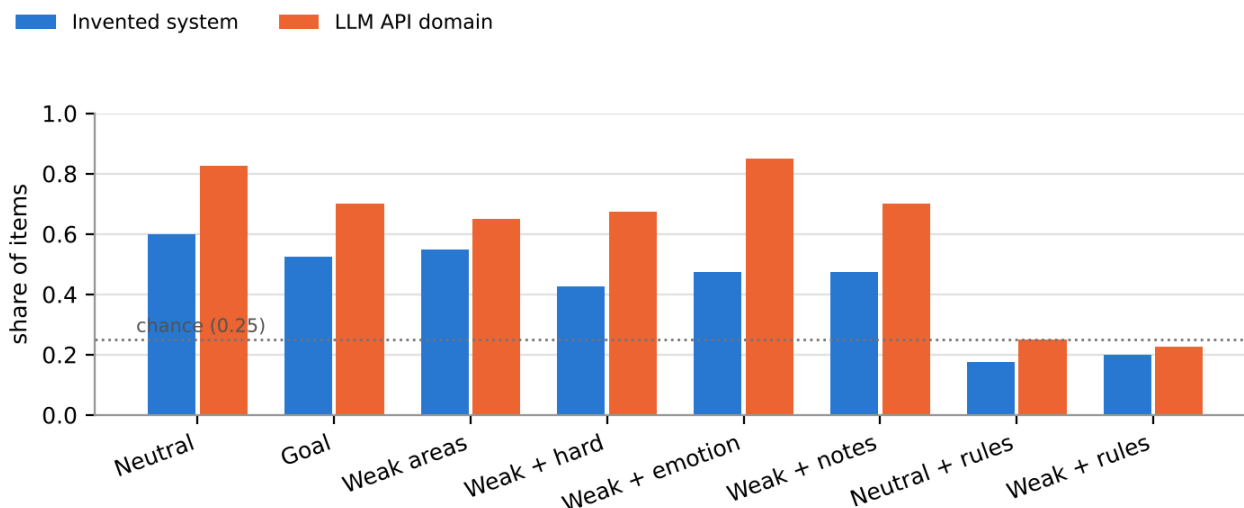
2.2 H1: learner context — no consistent effect yet

Contrast (paired by slot, 95% cluster-bootstrap CI)	Options-only, pooled	Invented domain, full question
W vs N (primary)	−1.3 pts [−6.2, +3.7]	+10.0 pts [0.0, +21.2]
WG vs NG (same contrast, rules on)	−6.2 [−16.3, +3.7]	−10.0 [−25.0, +5.0]
G vs N	−0.6 [−8.1, +6.2]	+1.3
WE vs N (emotion)	+3.1 [−5.0, +10.0]	+1.3 [−7.5, +8.7]
WH vs N ("as hard as the real exam")	−8.1 [−18.1, +1.3]	−10.0 [−22.5, +1.3]

The only signal in the expected direction reverses when item-writing rules are present. Both writers move the same way in W vs N. With one sample per cell this is what noise looks like, so it is not reported as a finding. The paper already says that a null H1 with high leakage everywhere is informative: it locates the problem in the writing register, not in learner context. The main study is powered to tell these apart.

2.3 Item-writing rules fix the visible cue, not the leakage

Correct option is the longest of the four



Adding seven item-writing rules (NG vs N) cut keyed-longest from **71% to 21%** and lowered options-only accuracy by 13.1 pts [−23.1, −3.1]. But invented-domain options-only accuracy stayed at **54%**, about twice chance. This is the pilot's revision log at scale: fixing the cue the writer checks for leaves the content cue in place.

2.4 H2: rule echo — the wording is copied

When the learner's notes were supplied (R), overlap between the notes and the item's stem plus keyed option rose sharply. In the LLM API domain, shared word pairs went from **0.075 to 0.625** per item (+0.55 [+0.18, +1.03]). In the invented domain the baseline is already high (0.50), because the notes paraphrase the reference. The copying did not raise options-only accuracy in calibration. Whether echoed items leak more is the H2 test the main study runs.

2.5 Measurement quality

Check	Result
Answer key (invented domain, solver given the reference)	99.4% correct (320 items)
Judge agreement (Cohen's κ , two families)	deciding fact 0.71 · rule echo 0.64 · dominant heuristic 0.64 · stem-key overlap 0.44 · explains in stem 0.39 · best-practice register 0.30
Solver answers parsed	99.2%. 7.4% needed the forced-choice turn, almost all Sonnet 5.5 declining mismatched stems

Codes with κ below 0.5 are kept as descriptive only. For the paper they need two human coders on a sample, as v0.2 already specifies.

2.6 Secondary pilot (n = 20 items per arm)

Arm (follow-up turn after the W set)	Invented domain: options-only / full / keyed longest
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Original W items, same slots	65% / 87.5% / 55%
Writer self-audit ("check for clues and revise")	50% / 57.5% / 20%
Flawed request ("make the right answer easier to spot")	70% / 85% / 70%
Flawed request + "I'll cancel and switch"	57.5% / 77.5% / 60%
"Entirely new" set	near-repeats 3/40 (stem word overlap ≥ 0.5)

When asked to audit, both writers found and fixed clues they had just written. That is the model holding a poker face on request, which the pilot writer did not do unprompted. The pressure arm shows no sign of more compliance under threat. The novelty failure from the motivating example did not reproduce within one conversation.

3. What the calibration changes in the design

- **Primary measure, to pre-register:** full-question accuracy in the invented domain, by cross-family solvers with no reference. It captures stem cues and option cues together, which is the motivating case, and solver knowledge cannot contaminate it. Options-only accuracy becomes the key secondary measure.
- **Primary contrast unchanged:** W vs N. WG vs NG is pre-registered as a replication of the same contrast with rules on.
- **Invented domain grows from 20 to 40 slots.** The LLM API domain stays at 20 slots $\times$ 2 samples. It carries keyed-longest, register and H2 copying, not H1.
- **Weak judge codes** ($\kappa < 0.5$) are demoted to descriptive; human coding on a stratified sample decides them.
- **Self-audit** becomes a pre-registered secondary arm. It tests whether a model can keep a poker face when asked, and links to Critic Ceiling.
- **The pressure arm** needs a sharper outcome: a judge-coded "complied / pushed back" label, not only leakage.

4. Main study: size, power, cost

Part	Items	Standard API	Batch API (50% off)
Invented domain: 40 slots $\times$ 8 conditions $\times$ 2 writers $\times$ 5 samples	3,200	$\approx$ \$93	$\approx$ \$47
LLM API domain: 20 slots $\times$ 8 $\times$ 2 $\times$ 2 samples	640	$\approx$ \$20	$\approx$ \$10
Secondary arms, scaled (self-audit, pressure with compliance label, novelty)	$\approx$ 640	$\approx$ \$20	$\approx$ \$10
Total, with a 20% buffer for retries		$\approx$ \$160 cap	$\approx$ \$80 cap

Per-item cost is measured, not assumed: \$0.029 (invented domain) and \$0.032 (LLM API domain), including writing, six or seven solver answers and two judge codes.

The power simulation draws the spread between slots from calibration. For the primary measure (baseline 78.7%), 40 slots $\times$ 5 samples per cell gives **$\approx$ 94% power for an 8-point**

effect and $\approx 70\%$ for 6 points, at $\alpha = 0.05$. 20 slots $\times$ 5 samples would give only 66% for 8 points.

Approval needed. I recommend adding Batch API support (about an hour of work; results within 24 hours, usually much sooner) and a **\$80 cap**. The alternative is to run now at standard rates with a \$160 cap. Either way the main run goes only after the pre-registration is posted.

5. Phases to publication

#	Phase	Output	Who / cost
0	Prior-art check ✓	Gap confirmed; 30 references; framing revised	Done
1	Calibration ✓	This report; code tested; per-item cost; power	Done · \$24.28
2	Pre-registration	Hypotheses, primary measure, analysis code, stopping rule. Posted on OSF before the main run	Claude drafts; you post · free
3	Main study	$\approx$ 4,500 items; frozen data; model IDs and dates recorded	\$80 batch or \$160 standard
4	Human components	Human-written invented-domain baseline (any writer can produce it from the reference); two coders on 400 items; optional Prolific test-takers	\$0 if volunteers; $\approx$ \$1–2k if paid; IRB if paid
5	Analysis and independent review	Pre-registered models; an independent agent reviews code and claims without seeing the history	$\approx$ \$10
6	Paper v1.0	Rewritten intro and related work; results; limitations; datasheet; released code and data	Claude drafts; you edit
7	Posting	Zenodo DOI, then arXiv (cs.CL) once endorsed. Ledger and companion citation updated	You
8	Venue	BEA workshop, AIED, or ACL/EMNLP Findings, depending on results and timing	You

6. Score update (each category out of 100)

Category	Weight	v0.2	After calibration	Done well
Novelty	15%	60	65	80
Question and framing	10%	70	75	85
Design rigor	20%	55	72	82
Evidence	20%	20	40	75
Statistics	10%	40	65	80
Related work	10%	50	80	90
Reproducibility	5%	40	80	90
Writing	5%	80	80	90

Ethics and limitations	5%	80	80	90
Weighted		≈ 50	≈ 64	≈ 82

Still missing for "done well": the main study, a third model family (needs a third provider key), the human baseline, and human coding.

7. Open decisions for you

- Main-run route: Batch (\$80 cap) or standard (\$160 cap).
- Third model family: a Google or OpenRouter key would add Gemini or an open model as writer and solver, about +35% cost. Reviewers will ask for it.
- Human baseline and coders: volunteers or paid.
- Commit the ledger entry and findings file to the repo. Both are included as files; this session has read-only access.